

Information Retrieval – Viva Questions (Sem VI)

Practical 1: Document Indexing and Retrieval

Q1. What is Information Retrieval (IR)?

Answer: Information Retrieval is the process of finding relevant documents or information from a large collection based on a user query.

Q2. What is document indexing?

Answer: Document indexing is the process of organizing documents so that they can be searched quickly.

Q3. What is an inverted index?

Answer: An inverted index is a data structure that stores words as keys and the list of documents containing those words as values.

Q4. Why is tokenization used in IR systems?

Answer: Tokenization splits text into smaller units (tokens or words) so they can be processed and indexed.

Q5. What is a retrieval system?

Answer: A retrieval system searches the index and returns documents relevant to the user's query.

Practical 2: Retrieval Models

Boolean Retrieval Model

Q1. What is a retrieval model?

Answer: A retrieval model defines how documents are matched with a user query.

Q2. What is the Boolean retrieval model?

Answer: It retrieves documents using logical operators like AND, OR, and NOT.

Q3. What is a term-document matrix?

Answer: It is a table showing which terms appear in which documents.

Q4. What does the AND operator do in Boolean queries?

Answer: It retrieves documents containing all the specified terms.

Q5. What does the OR operator do?

Answer: It retrieves documents containing at least one of the terms.

Vector Space Model

Q6. What is the Vector Space Model?

Answer: It represents documents and queries as vectors and calculates similarity between them.

Q7. What is TF-IDF?

Answer: TF-IDF stands for Term Frequency–Inverse Document Frequency. It measures the importance of a word in a document.

Q8. What is cosine similarity?

Answer: Cosine similarity measures how similar two vectors are by calculating the cosine of the angle between them.

Q9. Why is TF-IDF better than simple word count?

Answer: Because it reduces the importance of very common words.

Practical 3: Spelling Correction in IR Systems

Q1. What is spelling correction in IR?

Answer: It detects and corrects misspelled words in user queries.

Q2. What is edit distance?

Answer: Edit distance is the minimum number of operations needed to convert one word into another.

Q3. What operations are used in edit distance?

Answer: Insertion, deletion, and substitution.

Q4. What is Levenshtein distance?

Answer: It is a popular algorithm used to calculate edit distance.

Q5. Why is spelling correction important in search systems?

Answer: It improves search accuracy when users type incorrect words.

Practical 4: Evaluation Metrics for IR Systems

Q1. Why do we evaluate IR systems?

Answer: To measure how effectively the system retrieves relevant documents.

Q2. What is Precision?

Answer: Precision is the proportion of retrieved documents that are relevant.

Formula:

Precision = Relevant Retrieved / Total Retrieved

Q3. What is Recall?

Answer: Recall is the proportion of relevant documents that are retrieved.

Formula:

Recall = Relevant Retrieved / Total Relevant

Q4. What is F-measure?

Answer: F-measure is the harmonic mean of precision and recall.

Q5. What is Average Precision?

Answer: It measures the average precision score across different recall levels.

Practical 5: Text Categorization**Q1. What is text categorization?**

Answer: It is the process of assigning predefined categories to text documents.

Q2. What is a classifier?

Answer: A classifier is a model that predicts the category of a document.

Q3. What is Naive Bayes?

Answer: Naive Bayes is a probabilistic classification algorithm based on Bayes' theorem.

Q4. Why is it called "Naive"?

Answer: Because it assumes that features are independent of each other.

Q5. What is training data?

Answer: Training data is labeled data used to train the classifier.

Practical 6: Clustering for Information Retrieval**Q1. What is clustering?**

Answer: Clustering groups similar documents together without predefined labels.

Q2. What is K-Means clustering?

Answer: It is an algorithm that divides data into K clusters based on similarity.

Q3. What is a centroid?

Answer: A centroid is the center point of a cluster.

Q4. What is the silhouette score?

Answer: It measures how well objects fit within their clusters.

Q5. What is the difference between clustering and classification?

Answer: Classification uses labeled data, while clustering does not.

Practical 7: Web Crawling and Indexing

Q1. What is a web crawler?

Answer: A web crawler is a program that automatically visits and collects web pages.

Q2. What is robots.txt?

Answer: It is a file that tells crawlers which pages they are allowed to access.

Q3. Why is crawling delay used?

Answer: To avoid overloading web servers.

Q4. What is indexing in web search?

Answer: It stores processed web page information for faster searching.

Q5. What is dynamic content?

Answer: Content that changes automatically based on user actions or scripts.

Practical 8: Link Analysis and PageRank

Q1. What is link analysis?

Answer: It analyzes links between web pages to determine importance.

Q2. What is PageRank?

Answer: PageRank is an algorithm that ranks web pages based on the number and quality of links.

Q3. Who developed the PageRank algorithm?

Answer: It was developed by **Larry Page and Sergey Brin**, founders of Google.

Q4. What is the damping factor in PageRank?

Answer: It represents the probability of continuing to follow links (usually 0.85).

Q5. Why is PageRank important?

Answer: It helps search engines rank more important pages higher.